EXP 2 – Text Summarization

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The Basics of Blockchain Technology

Introduction

Blockchain technology is a revolutionary digital system that allows secure, transparent, and decentralized storage of data. It was first introduced in 2008 as the underlying technology for Bitcoin by the anonymous developer Satoshi Nakamoto. Since then, blockchain has evolved beyond cryptocurrencies and is now used in many industries such as finance, healthcare, supply chain management, and digital identity systems.

What is Blockchain?

A blockchain is a distributed digital ledger that records transactions across multiple computers in a network. Instead of storing data in a single centralized database, blockchain distributes copies of the ledger to all participants in the network. Each record in the ledger is stored in a block, and these blocks are connected in chronological order to form a chain.

Once a block is added to the blockchain, the information stored in it becomes extremely difficult to change. This immutability ensures that records remain secure and trustworthy.

Key Components of Blockchain

1. Blocks A block is a unit that stores a group of transactions. Each block contains three main elements:

Transaction data

A timestamp

A cryptographic hash of the previous block

This linking of blocks ensures that the chain remains secure and tamper-resistant.

2. Hashing Hashing is a cryptographic process that converts input data into a fixed-length string called a hash. Even a small change in the input will produce a completely different hash value. This property helps maintain the integrity of blockchain data.
3. Decentralization Traditional systems rely on a central authority such as a bank or government server. In blockchain, control is distributed among many computers called nodes. Each node maintains a copy of the ledger, ensuring transparency and reducing the risk of data manipulation.
4. Consensus Mechanism Before a new block is added to the chain, the network participants must agree on its validity through a consensus algorithm. Popular consensus mechanisms include Proof of Work (PoW) and Proof of Stake (PoS).

How Blockchain Works ?

When a transaction occurs, it is first broadcast to the network of nodes. These nodes verify the transaction using consensus rules. Once validated, the transaction is grouped with others into a block. The block is then added to the blockchain and shared across the network, updating all copies of the ledger simultaneously.

This process ensures transparency and prevents unauthorized modifications.

Applications of Blockchain:

Blockchain technology is being applied in several sectors:

Cryptocurrency: Digital currencies like Bitcoin use blockchain to record transactions securely.

Supply Chain Management: Companies track products from manufacturing to delivery.

Healthcare: Patient records can be stored securely and accessed by authorized parties.

Smart Contracts: Self-executing agreements stored on the blockchain automatically perform actions when conditions are met.

Advantages of Blockchain

Blockchain provides several benefits:

High security through cryptography

Transparency and traceability of transactions

Reduced need for intermediaries

Improved efficiency and trust in digital systems

Conclusion

Blockchain technology is transforming the way data and transactions are managed in the digital world. By combining decentralization, cryptographic security, and transparency, it offers a reliable system for secure information sharing. As the technology continues to develop, blockchain is expected to play a major role in shaping the future of digital infrastructure and global business systems.

Text Summarization:

Zero Shot Prompting:

Prompt:

You just need to summarize this article. And give me the Accuracy, Coherence, Simplicity, Speed, User, experience of your work in numbers.

Response:

ChatGPT:

Summary of The Basics of Blockchain Technology

Blockchain technology is a digital system used to store data securely, transparently, and in a decentralized manner. It was introduced in 2008 as the core technology behind Bitcoin by Satoshi Nakamoto. Over time, blockchain has expanded beyond cryptocurrency and is now used in industries such as finance, healthcare, and supply chain management.

A blockchain is a distributed digital ledger that records transactions across multiple computers in a network. Instead of storing data in one central database, the ledger is shared among many participants called nodes. Information is stored in units called blocks, and these blocks are linked together in chronological order to form a chain. Once data is recorded in a block and added to the blockchain, it becomes very difficult to modify, ensuring data integrity and trust.

The main components of blockchain include blocks, hashing, decentralization, and consensus mechanisms. Blocks store transaction data, timestamps, and the hash of the previous block, which connects the chain securely. Hashing is a cryptographic method that converts data into a unique fixed-length value, helping maintain data security. Decentralization removes the need for a central authority, while consensus mechanisms such as Proof of Work (PoW) and Proof of Stake (PoS) allow network participants to agree on the validity of transactions.

When a transaction occurs, it is broadcast to the network where nodes verify it using consensus rules. Valid transactions are grouped into blocks and added to the blockchain. This process updates all copies of the ledger and ensures transparency while preventing unauthorized changes.

Blockchain has several applications, including cryptocurrencies, supply chain tracking, secure healthcare records, and smart contracts that automatically execute agreements when conditions are met. Its advantages include strong security, transparency, reduced intermediaries, and improved efficiency.

Overall, blockchain technology provides a reliable and secure way to manage digital transactions and information. As it continues to develop, it is expected to play an important role in many future digital systems.

Gemini

The Basics of Blockchain Technology

Blockchain is a decentralized, distributed ledger technology that securely records transactions across a network of computers. Originally created for Bitcoin, it has expanded into various industries due to its unique structure and security features.

Core Mechanics

Structure: Data is grouped into Blocks, which contain transaction details, a timestamp, and a unique cryptographic hash of the previous block, forming an unbreakable chain.

Security: Through Hashing, any attempt to alter data changes the hash, alerting the network.

Governance: It operates without a central authority. Instead, it uses Consensus Mechanisms (like Proof of Work) where network nodes must agree on the validity of data before it is added.

how blockchain works diagram, AI generated

Getty Images

Impact and Benefits

Applications: Beyond cryptocurrency, it is used for Smart Contracts, transparent Supply Chain tracking, and secure Healthcare records.

Value Proposition: It eliminates the need for middle-men (like banks), ensures immutability (data cannot be changed), and provides a transparent audit trail for all participants.

Claude

Blockchain is a distributed digital ledger that records transactions across a network of computers rather than a single central database. Introduced in 2008 as the backbone of Bitcoin, it stores data in linked "blocks" secured by cryptographic hashing, making records nearly impossible to alter. The system operates without a central authority — instead, all participating nodes hold copies of the ledger and must reach consensus before any new block is added. This architecture makes blockchain highly secure, transparent, and tamper resistant. Beyond cryptocurrency, it is now applied in supply chain tracking, healthcare records, and smart contracts, with its core strengths being strong security, reduced reliance on intermediaries, and improved trust in digital transactions.

Few Shot Prompting:

Prompt:

You just need to summarize this article. And give me the Accuracy, Coherence, Simplicity, Speed, User, experience of your work in numbers.

Ex: 1) Blockchain is a distributed digital ledger that records transactions across a network of computers rather than a single central database. Introduced in 2008 as the backbone of Bitcoin, it stores data in linked "blocks" secured by cryptographic hashing, making records nearly impossible to alter. The system operates without a central authority — instead, all participating nodes hold copies of the ledger and must reach consensus before any new block is added. This architecture makes blockchain highly secure, transparent, and tamper-resistant. Beyond cryptocurrency, it is now applied in supply chain tracking, healthcare records, and smart contracts, with its core strengths being strong security, reduced reliance on intermediaries, and improved trust in digital transactions.

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Response:

ChatGPT

Blockchain is a decentralized and distributed digital ledger technology that securely records transactions across a network of computers instead of a single central database. It was introduced in 2008 as the core technology behind Bitcoin by Satoshi

Nakamoto. Data in blockchain is stored in blocks, which contain transaction details, timestamps, and cryptographic hashes linking them to previous blocks, forming a secure and chronological chain. This structure ensures immutability, meaning once data is added, it is extremely difficult to modify.

Blockchain operates without a central authority and relies on decentralized nodes that maintain copies of the ledger. Transactions are verified through consensus mechanisms such as Proof of Work (PoW) or Proof of Stake (PoS) before new blocks are added. Because of its transparency, security, and tamper-resistant nature, blockchain technology is widely used beyond cryptocurrency in areas such as supply chain management, healthcare data management, and smart contracts. Its main advantages include strong security through cryptography, reduced reliance on intermediaries, transparency of transactions, and improved efficiency in digital systems.

Gemini

The Basics of Blockchain Technology

Blockchain is a decentralized, distributed ledger that records transactions across a network of computers rather than a single central server. Introduced in 2008 as the foundation for Bitcoin, it has since expanded into healthcare, finance, and supply chain management.

How It Functions

The Chain: Data is stored in blocks containing transaction details, a timestamp, and a unique cryptographic hash of the previous block. This creates a chronological, tamper-resistant chain.

Security: Using cryptographic hashing, any change to a block’s data instantly changes its hash, breaking the connection to the rest of the chain and alerting the network to fraud.

Consensus: Instead of a bank or government, the network uses nodes (computers) that must agree on the validity of new data through a consensus mechanism (like Proof of Work) before it is added.

Why It Matters

By removing intermediaries, blockchain provides high transparency, immutability (data cannot be changed), and security. This allows for "Smart Contracts"—agreements that execute themselves automatically— and highly efficient tracking of goods or patient records.

Claude

Blockchain is a decentralized digital ledger introduced in 2008 for Bitcoin that records transactions across a network of computers instead of a single central database. Data is stored in blocks — each containing transaction details, a timestamp, and a cryptographic hash linking it to the previous block — making the chain tamper-resistant. No central authority governs it; instead, all network nodes must reach consensus before any new block is added. Today it powers far more than cryptocurrency, enabling supply chain tracking, secure healthcare records, and self-executing smart contracts, while eliminating the need for intermediaries and ensuring full transparency for all participants. Blockchain is a decentralized digital ledger introduced in 2008 for Bitcoin that records transactions across a network of computers instead of a single central database. Data is stored in blocks — each containing transaction details, a timestamp, and a cryptographic hash linking it to the previous block — making the chain tamper-resistant. No central authority governs it; instead, all network nodes must reach consensus before any new block is added. Today it powers far more than cryptocurrency, enabling supply chain tracking, secure healthcare records, and self-executing smart contracts, while eliminating the need for intermediaries and ensuring full transparency for all participants.

Chain of Thought Prompting:

Prompt:

Read the article and think step by step about the main topic, key points, and conclusions before writing a summary in 100-120 words.

Response:

ChatGPT

Blockchain is a decentralized digital ledger technology that securely records transactions across a network of computers instead of a central database. It was introduced in 2008 as the core technology behind Bitcoin by Satoshi Nakamoto. Data in blockchain is stored in blocks that contain transaction details, timestamps, and cryptographic hashes linking them to previous blocks, forming a secure chain. Once recorded, the data becomes nearly impossible to modify, ensuring integrity and trust. Transactions are verified by network nodes through consensus mechanisms such as Proof of Work or Proof of Stake. Due to its transparency, security, and decentralization, blockchain is widely used in cryptocurrency, supply chain tracking, healthcare data management, and smart contracts.

Gemini

Blockchain is a decentralized digital ledger that ensures secure, transparent data storage by distributing records across a global network of nodes rather than a single authority. Its structural integrity relies on blocks—containing data, timestamps, and cryptographic hashes—linked chronologically to create an immutable chain.

Key features include hashing for data integrity and consensus mechanisms (like PoW or PoS) to validate transactions without intermediaries. While born from Bitcoin, the technology now powers diverse sectors including supply chain, healthcare, and smart contracts. Ultimately, blockchain's shift away from centralized control fosters a more efficient, trustworthy digital infrastructure, positioning it as a cornerstone for future global business systems and secure information sharing.

Claude

Blockchain is a decentralized digital ledger that records transactions across a network of computers rather than in a single central database. Introduced in 2008 by Satoshi Nakamoto as the backbone of Bitcoin, it has since expanded into finance, healthcare, supply chain, and smart contracts. Its core components — blocks, cryptographic hashing, decentralization, and consensus mechanisms — work together to ensure data immutability and security. Transactions are verified by network nodes, grouped into blocks, and permanently linked in chronological order. The technology eliminates intermediaries, enhances transparency, and reduces the risk of tampering. As adoption grows across industries, blockchain is poised to become a foundational pillar of global digital infrastructure.

Role-Based Prompting:

Prompt:

As a Blockchain Tech Expert, Summarize this article in 100-120 words.

Response:

ChatGPT

Blockchain is a decentralized digital ledger technology that securely records transactions across a distributed network of computers. Introduced in 2008 as the underlying technology for Bitcoin by Satoshi Nakamoto, it stores data in interconnected blocks containing transaction details, timestamps, and cryptographic hashes linking each block to the previous one. This structure ensures immutability and strong data integrity. Transactions are verified by network nodes through consensus mechanisms such as Proof of Work (PoW) or Proof of Stake (PoS), eliminating the need for central authorities. Due to its transparency, security, and decentralization, blockchain is widely applied in cryptocurrency systems, supply chain tracking, healthcare record management, and smart contracts, improving trust, efficiency, and reliability in digital transactions.

Gemini

Blockchain is a decentralized, distributed ledger that records data across a global network of nodes, eliminating reliance on central authorities. Each block contains validated transaction data, a timestamp, and a cryptographic hash of the previous block, creating an immutable, chronological chain.

Security is maintained through hashing and consensus mechanisms (like PoW or PoS), which ensure all participants agree on the ledger's state. Beyond its Bitcoin origins, blockchain now optimizes sectors like healthcare and supply chains through smart contracts and transparent traceability. By prioritizing cryptographic integrity and decentralization, blockchain serves as a high-security framework that reduces intermediaries while enhancing digital trust and operational efficiency across modern global infrastructures.

Claude

Blockchain is a distributed ledger technology that records transactions across decentralized nodes, eliminating reliance on central authorities. Pioneered in 2008 by Satoshi Nakamoto for Bitcoin, it has since permeated finance, healthcare, supply chain, and smart contract ecosystems. Each block encapsulates transaction data, a timestamp, and a cryptographic hash of the preceding block — creating an immutable, tamper-resistant chain. Consensus mechanisms such as Proof of Work and Proof of Stake govern block validation across the network. Its architecture delivers cryptographic security, end-to-end traceability, and disintermediation at scale. As enterprise adoption accelerates, blockchain is rapidly becoming critical infrastructure for trust less, transparent, and efficient digital transactions globally.

AI Tool	Prompting Technique	Accuracy	Coherence	Simplicity	Speed	User Experience
ChatGPT	Zero-Shot Prompt	9	9	9	10	9
ChatGPT	Few-Shot Prompt	10	9	9	10	9
ChatGPT	Chain-of-Thought	9	10	9	8	8
ChatGPT	Role-Based Prompt	9	9	8	10	10
Gemini	Zero-Shot Prompt	8	9	10	10	9
Gemini	Few-Shot Prompt	10	9	9	10	8
Gemini	Chain-of-Thought	9	10	9	9	8
Gemini	Role-Based Prompt	8	9	8	10	10
Claude	Zero-Shot Prompt	9	10	9	10	9
Claude	Few-Shot Prompt	10	9	8	10	9
Claude	Chain-of-Thought	9	10	9	8	8
Claude	Role-Based Prompt	9	10	8	10	10